gpeyt1-13 · Compaction production

GEO PE · Long Calculation Problems I — Topic 3: Construction Observation, QA/QC & Safety

Problem

A contractor will compact 18,000 compacted yd³ of embankment with a single smooth-drum vibratory roller. Drum compacted width per pass = 7.0 ft; average speed = 2.5 mph; compacted lift = 8 in.; job efficiency = 80% (48 min/hr); test strip shows 6 passes are needed for 95% RC. Shrinkage from bank to compacted is 15% and the haul fleet delivers 300 bank yd³/hr.

Most nearly, how many roller-hours are required?

A. 10 hr
B. 47 hr
C. 59 hr
D. 50 hr

Given equations / data

> Compacted yd³/hr = 16.3 · W · S · L · efficiency / n (W ft, S mph, L in.)

Worked solution

1. Rate = 16.3 × 7.0 × 2.5 × 8 × 0.8 / 6 = 304.3 compacted yd³/hr
2. Hours = 18,000/304.3 = 59 hr
3. Shrinkage only converts compacted to bank yd³ — not needed because quantity is already compacted

Final answer

59 hr

Why the other values are wrong

- **10 hr** — multiplied by n instead of dividing
- **47 hr** — omitted the efficiency factor
- **50 hr** — applied the shrinkage factor to compacted quantity

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §2.3.3.1 Production Rate for Soil Compaction (Ch. 2 Construction)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=v9reh5FJnUc (no creator values copied).

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